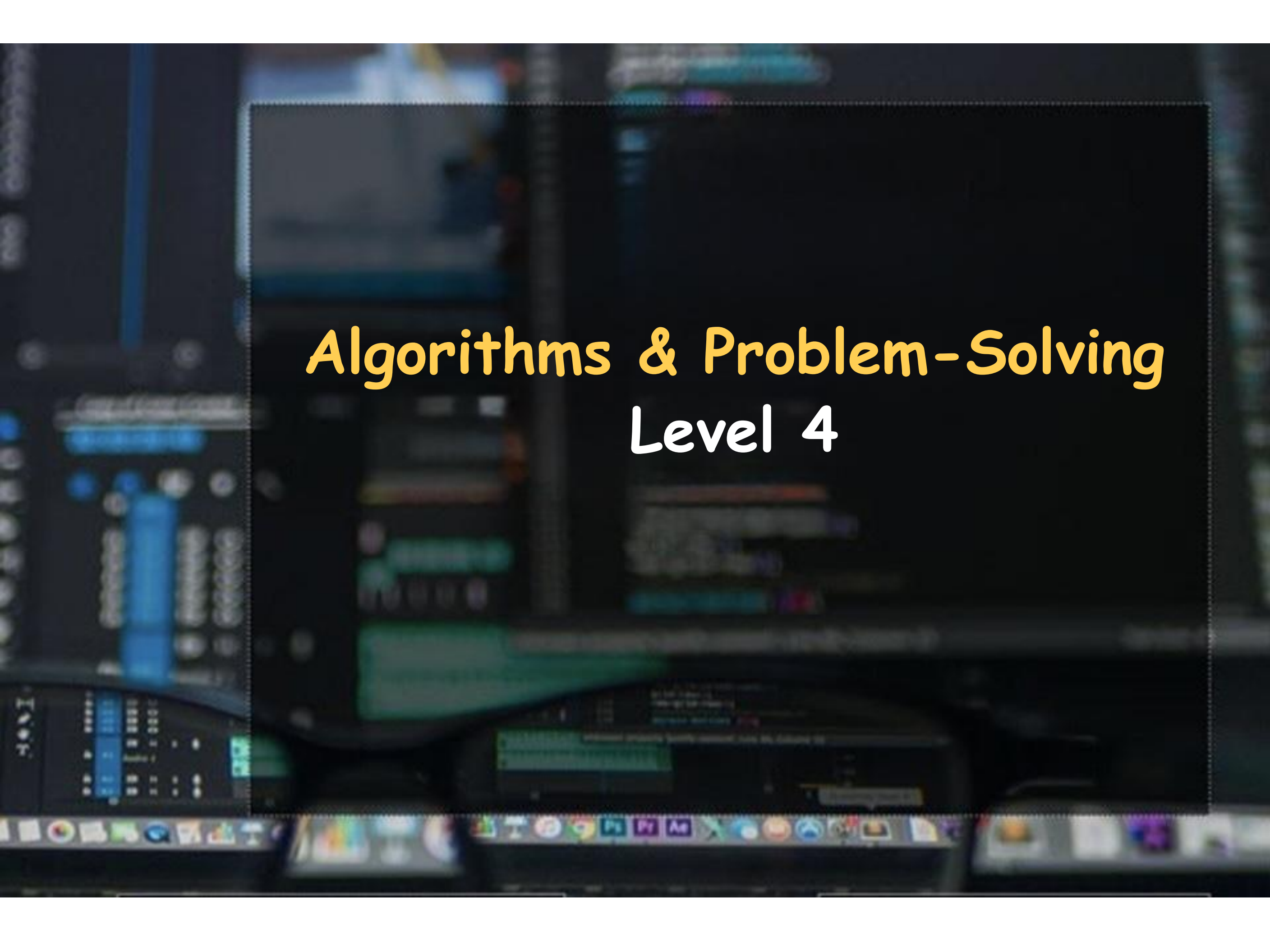
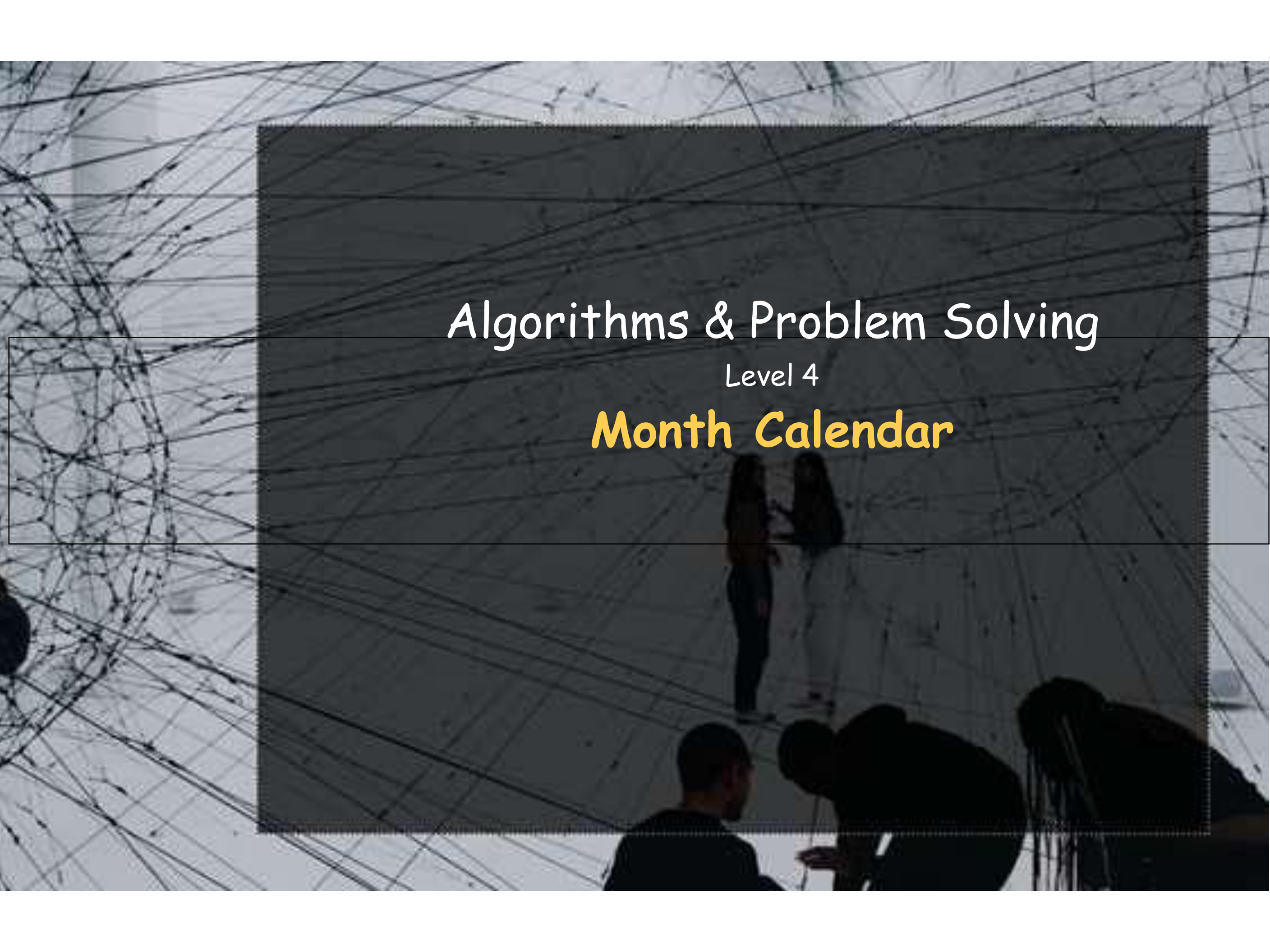




Algorithms & Problem-Solving Level 4

The background of the slide features a complex, abstract network of thin, dark lines crisscrossing over a lighter, textured surface. In the lower portion of the image, there are dark silhouettes of several people, some standing and some appearing to be in a discussion or collaborative work, adding a human element to the technical theme.

Algorithms & Problem Solving

Level 4

Month Calendar



Definition

- The goal is a program that prints a calendar for a specific month of a specific year, displaying the days in a coordinated manner starting from the correct day of the week (Sunday to Saturday).



Explanation

- The program is based on determining the day that the month begins on (e.g., Thursday or Sunday), then printing the days in order with the appropriate spaces before the first day.

Procedures

- 1 The ReadYear() function
 - Requires the user to enter the year.
- 2 The ReadMonth() function
 - Reads the month number from the user (from 1 to 12).
- 3 The IsLeapYear(int year) function
 - Checks if the year is a leap year:
 - A year is a leap year if:
 $(\text{year} \% 4 == 0 \ \&\& \ \text{year} \% 100 \neq 0) \ || \ (\text{year} \% 400 == 0)$
- 4 The NumberOfDaysInAMonth(int month, int year) function
 - Returns the number of days in the specified month, taking into account whether February has 28 or 29 days.
- 5 The DayOfWeekOrder(int day, int month, int year) function
 - Calculates the day of the week using the formula:
 $(\text{day} + y + (y/4) - (y/100) + (y/400) + ((31 * m) / 12)) \% 7;$
 - Output:
0 = Sun, 1 = Mon, 2 = Tue, ... etc.
- 6 The DayShortName(int dayOrder) function
 - Returns the shortened name of the day based on the following order:
"Sun", "Mon", "Tue", "Wed", "Thu", "Fri", "Sat"
- 7 The MonthShortName(int monthNumber) function
 - Returns the shortened name of the month, such as:
"Jan", "Feb", "Mar", ..., "Dec"

8 The PrintMonthCalendar(int month, int year) function

■ Displays the calendar as follows:

- Calculates the first day of the month
- Prints the month name and headings (Sun, Mon, Tue...)
- Allows spaces before the start of the month
- Prints the days in their correct order

Example:

Please enter a year to check? 2022

Please enter a Month to check? 9

Sep						
Sun	Mon	Tue	Wed	Thu	Fri	Sat
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	



Note:

- Counting starts on Sunday as the first day of the week.
- Formatting uses the printf function for accurate alignment.
- There is no month number validation, so entering a number outside of 1-12 will return 0 days.

Explanation:

- The equation used in `DayOfWeekOrder` is a modified Zeller's Congruence algorithm, which is an accurate method for determining the day of the week for any Gregorian date.

Summary

- ✓ The program uses the following concepts:
 - Modular functions
 - Mathematical calculation of the day
 - Rich printing using printf

Thank you 😊